exploratory_analysis

March 17, 2026

1 Milestone 1 Data Analysis

1.1 Comparison of Optimization Strategies for Logistic Regression

This notebook supports the Milestone report for the capstone project. It performs:

- basic exploratory data analysis on the Iris dataset
- data preprocessing and standardization
- training with BGD, SGD, and MBGD
- comparison of convergence, accuracy, and computational cost
- a small learning-rate sensitivity experiment

1.2 1. Imports

[1]:
```
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Make sure the project root is importable when this notebook is run from /
↪notebooks
PROJECT_ROOT = Path.cwd().resolve().parent
if str(PROJECT_ROOT) not in sys.path:
    sys.path.append(str(PROJECT_ROOT))

from data_loader import load_data
from train import train_model
from evaluation import compare_runs
from plots import plot_loss_curves, plot_accuracy_curves

plt.rcParams['figure.figsize'] = (8, 5)
```

1.3 2. Load the Iris dataset for EDA

[2]:
```
iris = load_iris(as_frame=True)
df = iris.frame.copy()
df['target_name'] = df['target'].map(dict(enumerate(iris.target_names)))

df.head()
```

[2]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target target_name
0       0      setosa
1       0      setosa
2       0      setosa
3       0      setosa
4       0      setosa
```

1.4 3. Basic dataset inspection

[3]:
```
print('Shape:', df.shape)
print('\nColumn names:')
print(df.columns.tolist())
print('\nMissing values per column:')
print(df.isnull().sum())
print('\nClass distribution:')
print(df['target_name'].value_counts())
```

```
Shape: (150, 6)

Column names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width
(cm)', 'target', 'target_name']

Missing values per column:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
target_name          0
dtype: int64

Class distribution:
```

```
target_name
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```

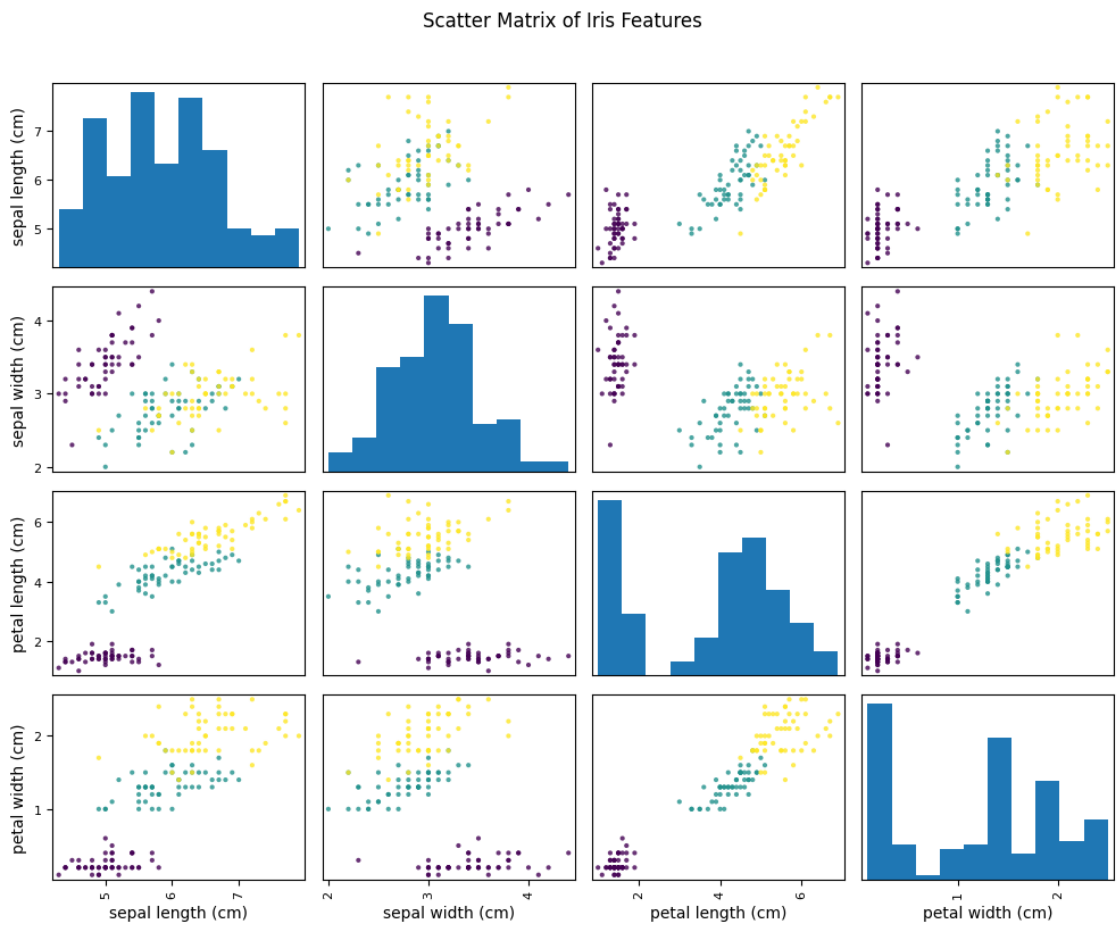
1.5 4. Summary statistics

[4]:
```
df.describe()
```

[4]:
```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)      target
count        150.000000  150.000000
mean           1.199333    1.000000
std            0.762238    0.819232
min            0.100000    0.000000
25%            0.300000    0.000000
50%            1.300000    1.000000
75%            1.800000    2.000000
max            2.500000    2.000000
```

1.6 5. Pairwise visualization

[5]:
```
pd.plotting.scatter_matrix(
    df.iloc[:, :4],
    c=df['target'],
    figsize=(10, 8),
    diagonal='hist',
    alpha=0.8,
)
plt.suptitle('Scatter Matrix of Iris Features', y=1.02)
plt.tight_layout()
plt.show()
```

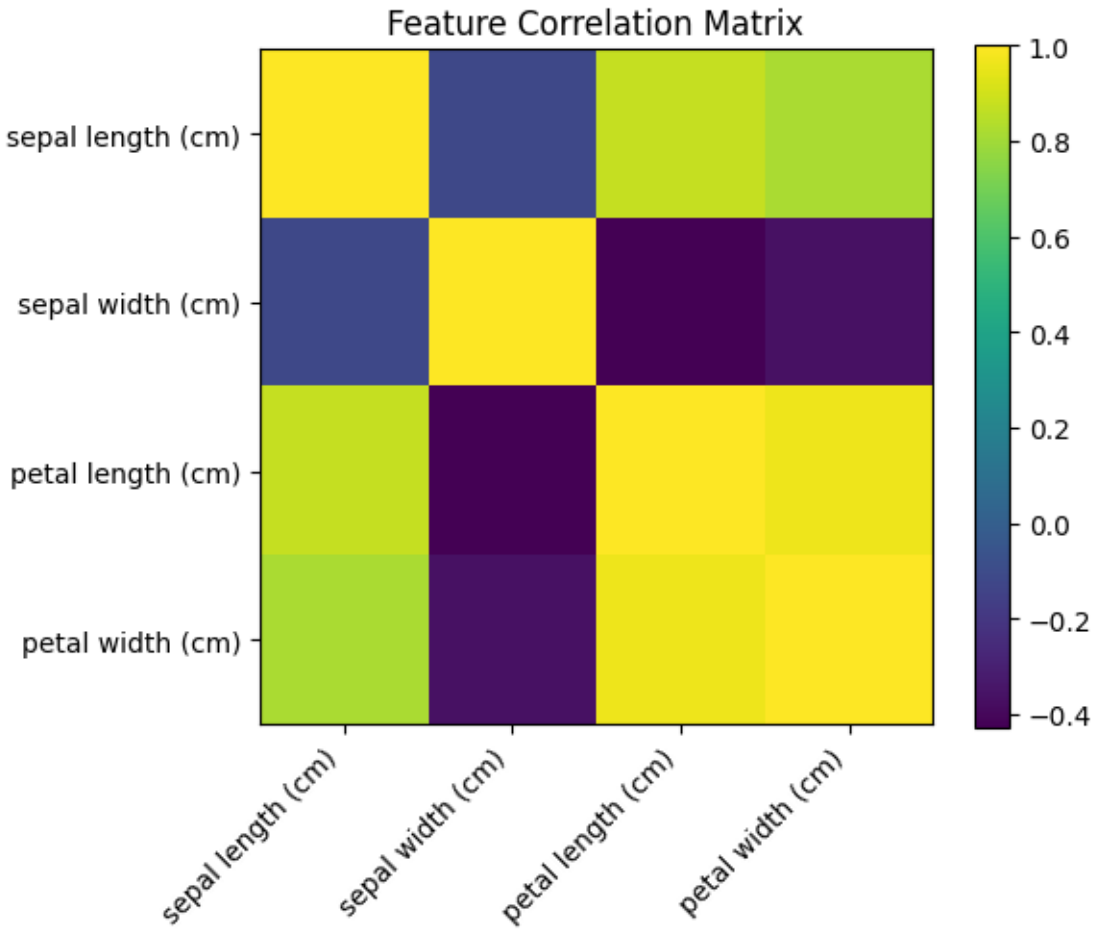


1.7 6. Correlation analysis

[6]:
```
corr = df.iloc[:, :4].corr()
corr
```

[6]:
```
                   sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)           1.000000         -0.117570           0.871754
sepal width (cm)           -0.117570          1.000000          -0.428440
petal length (cm)           0.871754         -0.428440           1.000000
petal width (cm)            0.817941         -0.366126           0.962865

                   petal width (cm)
sepal length (cm)          0.817941
sepal width (cm)          -0.366126
petal length (cm)          0.962865
petal width (cm)           1.000000
```

[7]:
```
plt.figure(figsize=(6, 5))
plt.imshow(corr, interpolation='nearest')
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45, ha='right')
plt.yticks(range(len(corr.index)), corr.index)
plt.colorbar()
plt.title('Feature Correlation Matrix')
plt.tight_layout()
plt.show()
```



1.8 7. Load standardized train/test data

[8]:
```
X_train, X_test, y_train, y_test = load_data(test_size=0.2, random_state=42)

print('Training shape:', X_train.shape)
print('Test shape:', X_test.shape)
print('Training class counts:', np.bincount(y_train))
print('Test class counts:', np.bincount(y_test))
```

```
Training shape: (120, 4)
Test shape: (30, 4)
Training class counts: [40 40 40]
Test class counts: [10 10 10]
```

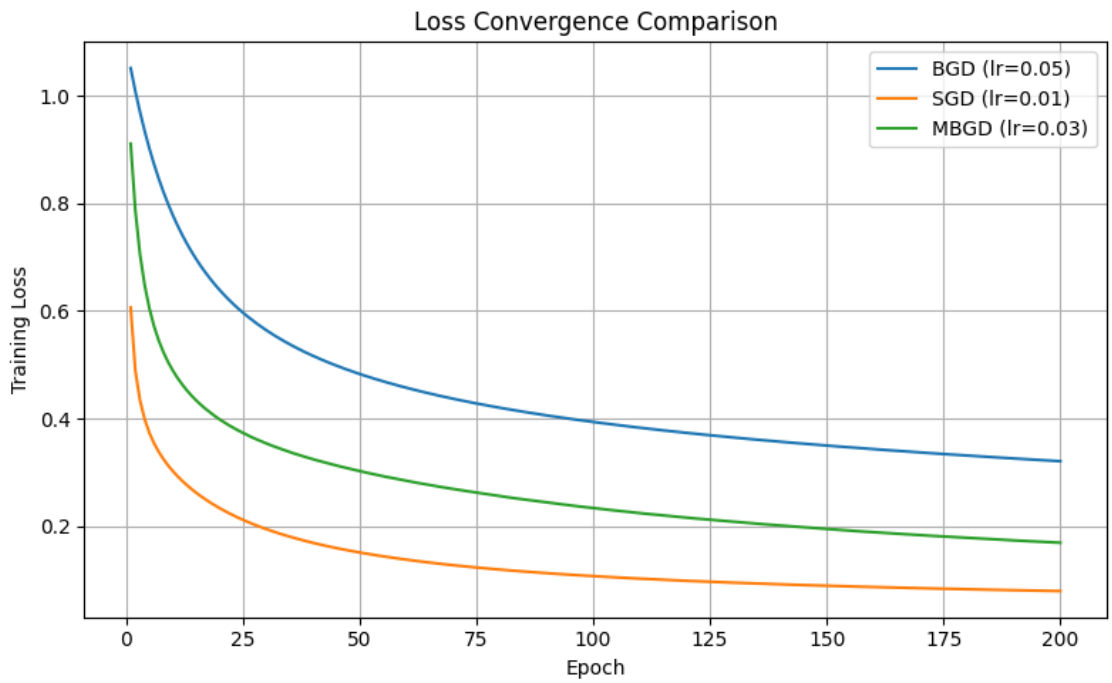
1.9 8. Main experiment: BGD vs SGD vs MBGD

[9]:
```
settings = [
    {'optimizer': 'bgd', 'learning_rate': 0.05, 'epochs': 200, 'batch_size':␣
 ↪16},
    {'optimizer': 'sgd', 'learning_rate': 0.01, 'epochs': 200, 'batch_size': 1},
    {'optimizer': 'mbgd', 'learning_rate': 0.03, 'epochs': 200, 'batch_size':␣
 ↪16},
]

histories = []

for cfg in settings:
    _, history = train_model(
        X_train,
        y_train,
        X_test,
        y_test,
        optimizer=cfg['optimizer'],
        learning_rate=cfg['learning_rate'],
        epochs=cfg['epochs'],
        batch_size=cfg['batch_size'],
    )
    histories.append(history)

summary_df = compare_runs(histories)
summary_df
```

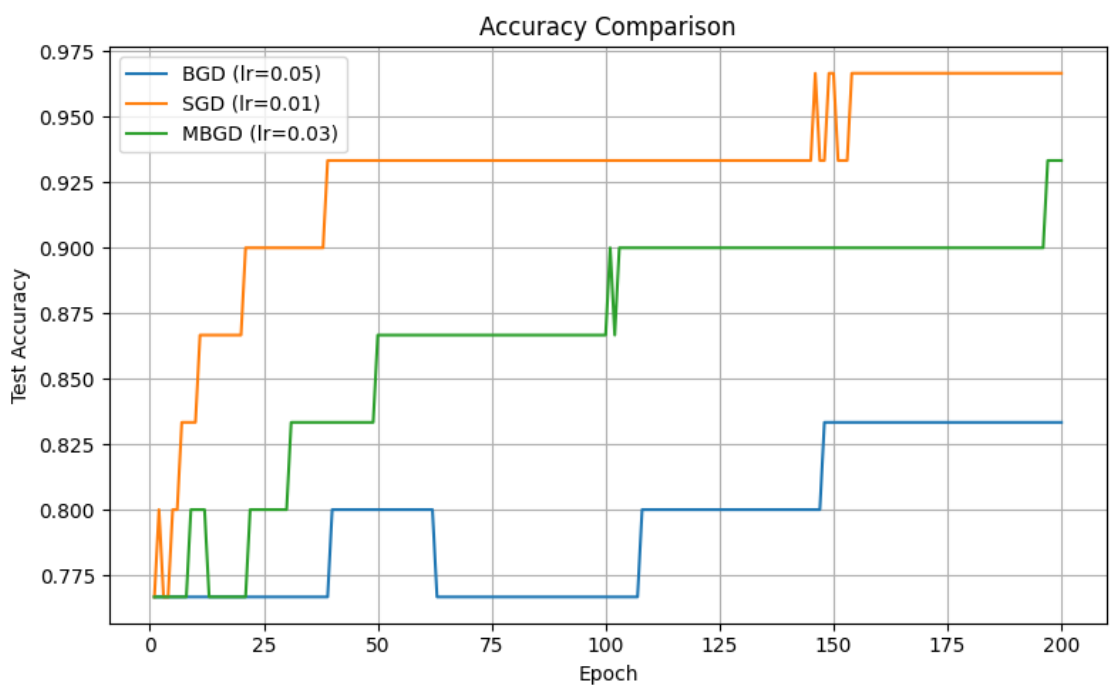
[9]:
```
  optimizer  learning_rate  final_train_loss  final_test_loss  \
0       bgd           0.05          0.320853         0.364136
1       sgd           0.01          0.079440         0.111994
2      mbgd           0.03          0.169326         0.214105

   final_train_acc  final_test_acc  avg_epoch_time  loss_std
0         0.925000        0.833333        0.000100  0.142892
1         0.975000        0.966667        0.001681  0.078096
2         0.958333        0.933333        0.000202  0.111898
```

1.10 9. Plot convergence and accuracy curves

[10]:
```
plot_loss_curves(histories)
```



[11]:
```
plot_accuracy_curves(histories)
```



1.11 10. Learning-rate sensitivity experiment

This section runs a small hyperparameter sweep to show how each optimizer reacts to different learning rates.

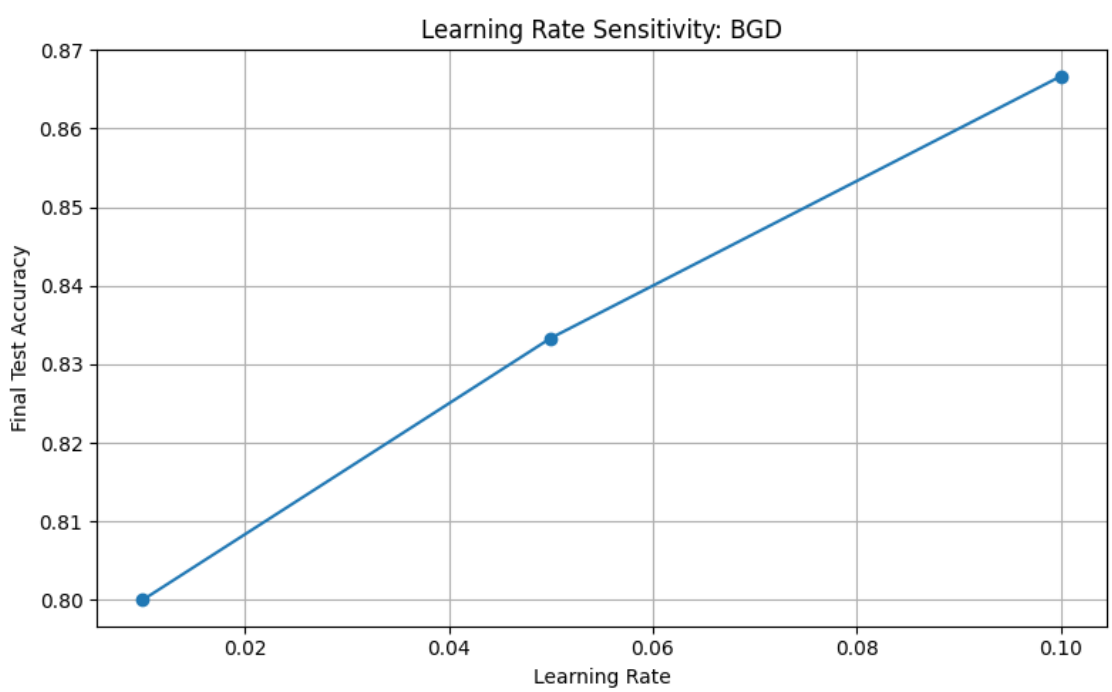
[12]:
```
lr_grid = {
    'bgd': [0.01, 0.05, 0.1],
    'sgd': [0.001, 0.01, 0.05],
    'mbgd': [0.01, 0.03, 0.05],
}

sweep_rows = []

for optimizer, lr_list in lr_grid.items():
    for lr in lr_list:
        _, history = train_model(
            X_train,
            y_train,
            X_test,
            y_test,
            optimizer=optimizer,
            learning_rate=lr,
            epochs=200,
            batch_size=16 if optimizer != 'sgd' else 1,
        )
        sweep_rows.append({
            'optimizer': optimizer,
            'learning_rate': lr,
            'final_train_loss': history['train_loss'][-1],
            'final_test_loss': history['test_loss'][-1],
            'final_train_acc': history['train_acc'][-1],
            'final_test_acc': history['test_acc'][-1],
            'avg_epoch_time': np.mean(history['epoch_time']),
        })

sweep_df = pd.DataFrame(sweep_rows)
sweep_df.sort_values(['optimizer', 'learning_rate']).reset_index(drop=True)
```

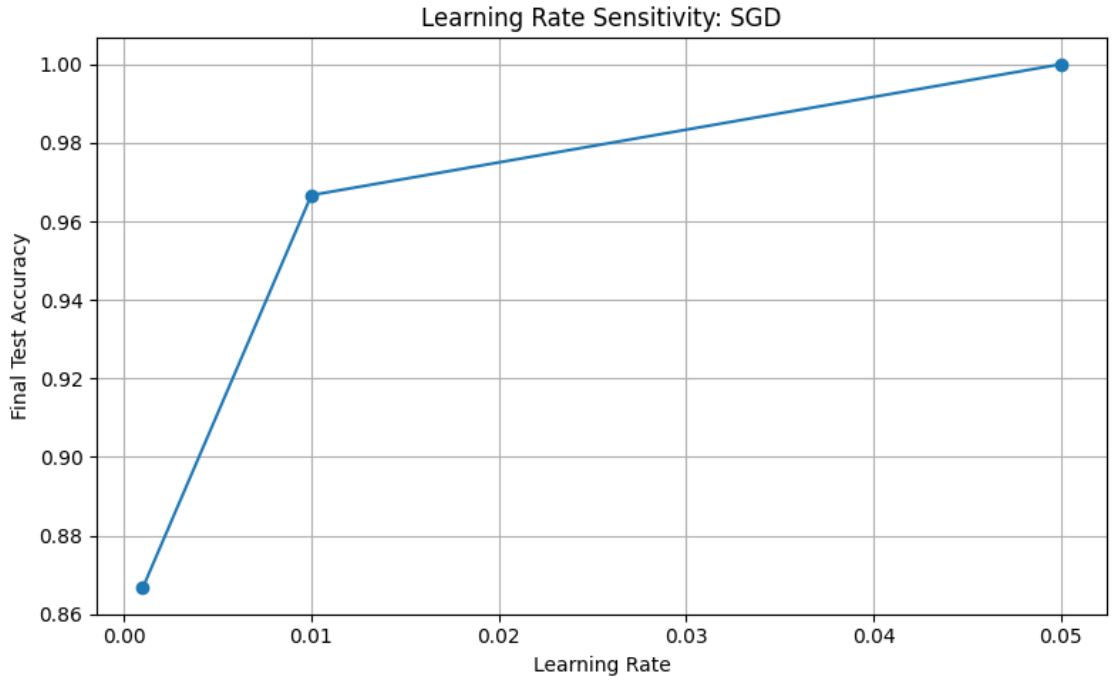
[12]:
```
  optimizer  learning_rate  final_train_loss  final_test_loss  \
0       bgd          0.010          0.518640         0.545970
1       bgd          0.050          0.320853         0.364136
2       bgd          0.100          0.251871         0.297899
3      mbgd          0.010          0.274479         0.319981
4      mbgd          0.030          0.169326         0.214105
5      mbgd          0.050          0.130785         0.172554
6       sgd          0.001          0.234178         0.280417
7       sgd          0.010          0.079440         0.111994
8       sgd          0.050          0.050110         0.066994
```

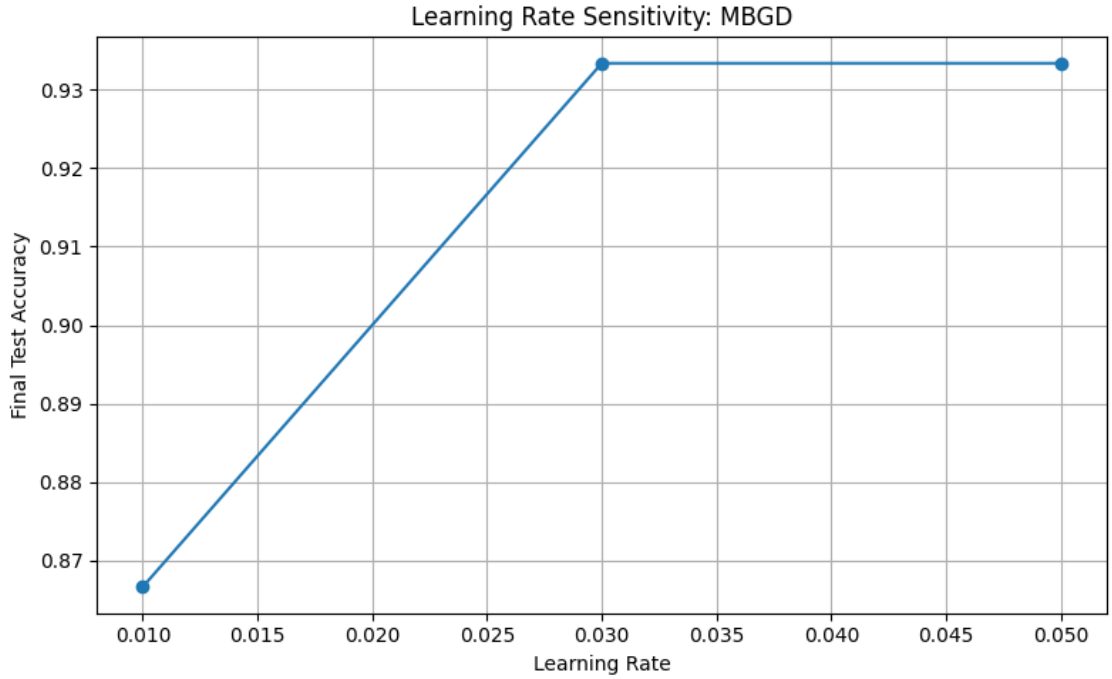
```
   final_train_acc  final_test_acc  avg_epoch_time
0         0.850000        0.800000        0.000111
1         0.925000        0.833333        0.000109
2         0.933333        0.866667        0.000101
3         0.925000        0.866667        0.000199
4         0.958333        0.933333        0.000198
5         0.966667        0.933333        0.000198
6         0.933333        0.866667        0.001652
7         0.975000        0.966667        0.001646
8         0.975000        1.000000        0.001642
```

1.12 11. Visualize test accuracy across learning rates

[13]:
```
for optimizer in ['bgd', 'sgd', 'mbgd']:
    subset = sweep_df[sweep_df['optimizer'] == optimizer].
 ↪sort_values('learning_rate')
    plt.figure()
    plt.plot(subset['learning_rate'], subset['final_test_acc'], marker='o')
    plt.xlabel('Learning Rate')
    plt.ylabel('Final Test Accuracy')
    plt.title(f'Learning Rate Sensitivity: {optimizer.upper()}')
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```




Learning Rate Sensitivity: SGD
Final Test Accuracy
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.00
0.01
0.02
0.03
0.04
0.05
Learning Rate


Learning Rate Sensitivity: MBGD
Final Test Accuracy
0.93
0.92
0.91
0.90
0.89
0.88
0.87
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
0.050
Learning Rate

1.13 12. Observations

- All three optimization methods converge on the Iris dataset.
- SGD often reaches lower loss faster in epoch-based comparison because it updates parameters more frequently.
- BGD tends to be smoother but slower.
- MBGD usually provides a compromise between computational efficiency and predictive performance.
- Learning rate strongly affects both convergence speed and final accuracy.